Revision Report / Response to Reviewers

Initial Placement for Fruchterman–Reingold Force Model With Coordinate Newton Direction

Hiroki Hamaguchi, Naoki Marumo, and Akiko Takeda

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Stephen Kobourov
Technical University Munich
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Dear Prof. Stephen Kobourov,

Thank you for directing us to the reviewers' comments and for giving us the opportunity to revise our manuscript. We confirm that we have received the review reports.

Based on the reports, we have carefully reviewed all comments and revised the manuscript accordingly. Our responses are provided below in a point-by-point manner. Where applicable, we refer to section, equation, figure, or algorithm numbers in the revised manuscript.

We hope the revised version is now suitable for publication and look forward to hearing from you.

Sincerely,

Hiroki Hamaguchi, Naoki Marumo, and Akiko Takeda

A Response to Reviewer A

We thank Reviewer A for the careful and constructive assessment of our manuscript. We appreciate the reviewer's observation that the proposed algorithm is interesting and that there are no major problems with the construction of the algorithm. We have revised the manuscript to address the concerns regarding the relationship between the proposed method and the Fruchterman–Reingold model, as well as the distinction between the FR and KK models.

A.1 Relationship Between the Proposed Approach and the FR Force Model

> First, the authors do not sufficiently consider how well their approach matches the FR force model.
>
> Originally, Fruchterman and Reingold [10] proposed the FR model and solved it by using a pseudo-physical simulation-based algorithm (by following Eades' spring embedder method [8]). By contrast, Kamada and Kawai [21] proposed the KK force model and solved it by a Newton-like method. Also, many succeeding studies followed these approaches, i.e., either by combining simulation-based algorithms with the FR model or by combining Newton-like methods with the KK model. Although Hosobe [19] showed that both models could be solved with a Newton-like method called L-BFGS, the results still suggested that the KK model was more suitable for L-BFGS than the FR model.
>
> However, this submitted paper adopts an unusual approach that combines a Newton-like method with the FR model. I understand that the paper is focused only on the

computation of initial layouts, but I think that the validation of adopting this approach is still insufficient.

It should be noted that the experimental results in Section 5 show that the proposed initial layout algorithm works better when it is combined with the L-BFGS algorithm for computing final layouts. It seems from Figure 6 that the combination of the proposed initial layout algorithm with the FR algorithm for computing final layouts does not work well except in limited cases like jagmesh1.

If the authors think that the proposed algorithm should be combined with the L-BFGS, they should adopt the KK model instead of the FR model, which would be more natural and more persuasive. If the authors think that the proposed algorithm works well for the FR layout algorithm (or another simulation-based algorithm), they should more deeply explore this direction and present clearer results in the paper.

Response: ご指摘に感謝いたします。原稿では確かになぜ我々が FR モデルに焦点を当てたのかを十分に正当化できていませんでした。

重要なのは次の 2 点です。まず、1 点目として提案手法は確かに KK モデルなどの対象についても適用可能であると考えており、その適用方法については新しい section を

A.2 Difference Between FR and KK Models

Second, the authors do not sufficiently consider differences between the FR force model and others, especially the KK model, which becomes apparent especially in Subsection 7.1.

Here the authors first say "Although we utilized the coordinate Newton direction only for the optimization problem (1), we can see that its application is not necessarily limited to the FR force model alone", and then present "objective functions arising from graphs". Does the authors mean that this discussion applies to other force models including the KK model? If so, I think that the authors should adopt the KK model instead of the FR model.

In addition, the authors present the graph isomorphism problem as an application of the proposed algorithm. I think that the KK model would be more suitable for this application than the FR model.

The following might work as hints for the authors' paper revision. Intuitively, the FR model exhibits soft behavior while the KK model exhibits hard behavior. Therefore, the FR model is usually used for dynamic graph layouts like animation and interaction while the KK model is usually used for static layouts. Also, the FR model is sometimes used for very large graphs because simulation-based algorithms for the FR model can be more scalable than Newton-like algorithms for the KK model. One plausible direction for the authors might be to combine the proposed algorithm with a scalable simulation-based algorithm for the FR model and to apply it to very large graphs.

Response: 示唆に富むご指摘に感謝いたします。大まかには、First の点で述べたことと重なる部分もありますが、一般化は可能であり、それを基に議論しました。

graph isomorphism への応用に関するご指摘にも同意いたします。この応用は FR モデルに特に自然なものとは言い難く、本論文の主たる貢献から読者の注意を逸らす可能性があるため、改訂稿ではこの議論を削除します。

一方で、FR モデルの soft behavior に関する査読者のご指摘は、本研究の位置づけを明確化する上で非常に有益であると考えています。改訂稿では、提案手法の重要な応用として、dynamic または animation-like なグラフ描画シナリオに関する議論を強化します。具体的には、動的グラフにおける各時刻のレイアウト更新を、変更後の FR objective に対する再最適化のための初期配置問題として捉えます。このような設定では、すでに安定したレイアウトが得られており、そこに少数の頂点や辺が追加される場合、あるいはレイアウトやグラフ構造に小さな摂動が加えられる場合に、提案する座標ごとの手法によって、改善された初期レイアウトを効率的に計算できます。この利用場面は FR モデルの soft behavior とよく整合しており、また査読者が示唆された scalability-oriented な方向性とも整合しています。したがって、この側面を示すために、大規模かつ animation-like なグラフ描画シナリオに関する新しい実験を追加します。

A.3 Notation

The following are minor comments.

- Symbol E is used for two different purposes, one to indicate a set of graph edges and the other to indicate energies between nodes possibly with subscripts and superscripts. It will be better to use different symbols for these different concepts.

- I wonder if $(i,j) \in V^2$ with $i \neq j$ in (1) is appropriately intended. This treats both (i,j) and (j,i) in the sum.

- Symbol ϵ appears only once in Subsection 4.4, which seems strange.

- Symbol f_i at line 6 of Algorithm 1 probably lacks the superscript "a".

- In Subsection 5.3, the word "L-BFGS" that appears as superscripts of N does not seem to be correctly specified. The hyphen mark looks too long.

- In the optimization problem called "objective functions arising from graphs" in Subsection 7.1, "=" probably should be ":=".

- The header of Subsection 8.1 seems to have a problem with the hyphen mark again.

Response: Thank you for pointing out these notation and typography issues. We addressed them as follows.

- We avoided the symbol conflict by using a dedicated symbol for the edge set (so that the energy terms $U_{i,j}$ are not confused with the edge set).
- We intended the sum to be over all ordered pairs of distinct vertices, so that both (i,j) and (j,i) are included. We clarified this in the text and notation.
- We clarified the role of ϵ as the minimum separation in the discrete point set Q and its use in preventing divergence.
- We corrected the missing superscript in Algorithm 1 (the update uses the attractive-only objective f_i^{a}).
- Regarding the reviewer's note on the "L-BFGS" superscripts: the hyphen was already intended as a proper hyphen in math mode, but its appearance can still be misleading. To avoid confusion, we now use the spelling "LBFGS" when it appears inside math expressions (e.g., in superscripts), while keeping the standard spelling "L-BFGS" in running text.
- We corrected the definition symbol in the "objective functions arising from graphs" formulation.
- We checked the subsection headers and ensured hyphenation uses proper LaTeX conventions.

B Response to Reviewer B

We thank Reviewer B for the careful review and constructive suggestions. We appreciate the positive assessment that the paper addresses a topic of clear interest to the JGAA audience, that the technical explanation is comprehensive, and that the mathematical treatment is mostly easy to follow. We have revised the manuscript to improve the framing of the contribution, the discussion of related work, the experimental evaluation, and the organization of the later sections.

B.1 Related Work on Multilevel Approaches

> In the Related Work section, in 3.1 Multilevel approaches are quickly dismissed. I don't understand the statement the statement about sfdp being "prior work" to FR - as the latter came way later than the former. This sentence should be clarified. Also, I believe that further space could be given to multi-level approaches, also considering that could be one potential application/extension of this technique beyond this paper.

Response: Thank you for pointing this out. We clarified the statement around sfdp: our intent was to reference it as prior work on scalable, multilevel force-directed graph drawing rather than implying chronological priority over the FR model. We also expanded the discussion in the related-work section and clarified how our method is complementary to multilevel pipelines (in particular, as an improvement to the refinement stage).

> Multilevel approaches such as sfdp in Graphviz are important prior work on scalable force-directed graph drawing. These methods use hierarchical coarsening and refinement to achieve efficient layout computation for large graphs.
>
> Our work is complementary to such multilevel methods. Optimization techniques such as L-BFGS can be directly applied to the refinement stages within multilevel pipelines. Since the approaches proposed in this study can be incorporated into multilevel frameworks with appropriate modifications, our primary objective is to quantify and enhance the refinement procedure itself. Therefore, we deliberately exclude coarsening and other approximation steps from the present analysis.

B.2 Uncertain Framing of FR Versus L-BFGS

> The framing of this paper is somewhat uncertain. Despite the paper title and general orientation towards FR, L-BFGS is consistently and repeatedly mentioned as a better alternative. Even in the related work section, in Section 3.1, only L-BFGS is mentioned as a ready-to-go measure for application in Multilevel approaches. In the experiments the paper explicitly mentions that FR is consistently outperformed by L-BFGS - then why we should use FR (even with the improved initial placement) at all? The paper does not make a good argument why we should choose that over the other, thus in a way making me question even the title of the paper. In general, it's unclear what's the impact of this research and how researchers could build upon it.

Response: FR を使うべきと思われる状況は、例えば、動的グラフのレイアウト更新のようなシナリオであると考えています。動的グラフでは、各時刻のレイアウト更新を、変更後の FR objective に対する再最適化のための初期配置問題として捉えることができます。reviwerA でも指摘されたように、FR モデルの soft behavior はこのようなシナリオとよく整合しており、また reviewerB が示唆

された scalability-oriented な方向性とも整合しています。したがって、この側面を示すために、大規模かつ animation-like なグラフ描画シナリオに関する新しい実験を追加します。

B.3 Expansion of the Evaluation Dataset

The evaluation yields positive results, but in my opinion it could be further improved. First, I would suggest to expand 5.2 it in terms of number of graphs. More graphs from the Florida library could be used, or another source could be [1]. This expanded datasets already includes 3elt, but expands also on grids, cylinders, and other regular graphs that make up a solid benchmark set, and will provide a wider set of instances to discuss. If not all figures can be included in the main body of the paper, these could be moved in the supplemental material.

[1] Bartel, Gereon, et al. "An experimental evaluation of multilevel layout methods." International Symposium on Graph Drawing. Berlin, Heidelberg: Springer Berlin Heidelberg, 2010.

Response: Thank you for the suggestion. 実験を追加します。(Appendix などで良さそう)

B.4 Visualization and Color Map

Concerning the visualization of the layout, I don't believe that the index colouring adds much. A rainbow map, however, it's not a great choice. If color should be kept, maybe I would consider using a sequential continuous color map with a legend (see [2] for examples).

[2] https://matplotlib.org/stable/users/explain/colors/colormaps.html

Response: Thank you for the suggestion. We agree that the current index-based coloring does not add much interpretive value. We revise the visualization by replacing the rainbow map with a sequential continuous color map (Since some graphs such as jagmesh1 have a symmetric structure, the color can help visually distinguish different vertices). We also clarified the color only represents the vertex index for visual distinction.

B.5 Effect of Initial Placement Over Time

One thing that comes out of the charts in Figure 6 is that this different initial positioning has limited effect on the way the objective function decreases over time - which is an interesting phenomenon. In FR, the chart appears shifted below by how "better" the initial positioning is, with the non-CN version never catching up. Differently, it seems that the objective functions of L-BFGS and CN-L-BFGS tend to converge before the iteration cap. Does this mean that the positive effect of the initial placement tends to fade faster with L-BFGS, despite the better initial condition?

Response: Yes, this interpretation is consistent with our observations. We added a brief explanation in the experimental discussion: for FR, a better initialization tends to yield a persistent offset in the objective curve, whereas for L-BFGS the curves often converge earlier so the relative advantage can diminish once near-stationary points are reached. We also clarified that the effect should be judged not only by the final objective value, but also by the early-stage convergence behavior and runtime.

B.6 Singularities in dwt_992 and collins_15NN

The dwt_992 and collins 15NN singularities could be described more. Would it be possible to generalize, highlighting what makes these two cases so different from each other?

Speaking of singularities, in the charts it seems like there is a spike in $f(X)$ right after the 0 seconds mark (like in 3elt, btree or cycle) is that an artifact?

Response: todo?

In some FR runs, a small increase in $f(X)$ appears immediately after the start of the iterations; this can happen because the first few displacement steps are relatively large before the temperature decreases, so it is an artifact of the early-stage update schedule rather than a failure of the initialization.

The effect of the initial placement also differs across the two solvers: for FR, a better initialization tends to produce a persistent offset in the objective curve, whereas for L-BFGS the curves often converge earlier so the relative advantage can fade near stationary points. Thus, the benefit of initialization should be judged not only by the final objective value, but also by the early-stage convergence behavior and runtime.

B.7 Organization of Sections 6 and 7

I am less persuaded from Sections 6 and 7. Section 6 includes considerations that I would include in Section 4, as to further justify your optimization design. I would also simplify the discussion, removing subsections and integrating their contents in Section 4. In Section 7 I would focus on the impact of this approach on graph drawing more specifically. For example, where is the next step for this technique, in the context of graph drawing? Will its presence in a drawing pipeline make a difference, or the gains in quality are not sufficient to justify its use? What about its inclusion in a multi-level pipeline (see also a similar point above)?

Response: これは応用例の section を ReviwerA で答えたのと被る。応用例を消すか、増やすか、記述をもっと精緻化するか。

B.8 Moving the Section 8 to the Appendix

I would move Section 8 to supplemental material out of the main body of the paper.

Response: Thank you. In the revision, we moved the implementation-oriented material (NetworkX implementation details) to the Appendix so that the main narrative focuses on the method, experiments, and discussion.

B.9 Bibliography

Bilbiography must be updated. Some papers do not include any venue, others do not have a DOI, others only report their archive version (while journal versions are available).

Response: これが断トツで自信がないが、対応が不完全。Thank you for the careful note. In the revision, we reviewed the bibliography entries and improved completeness where possible (e.g., adding missing metadata and cleaning local-only fields), and we checked for cases where a journal/conference version should be cited instead of an archive-only version.

Closing

We again thank the editor and the reviewers for their valuable comments. We believe that the revised manuscript is clearer in scope, better positioned with respect to existing force-directed graph drawing methods, and more explicit about both the strengths and limitations of the proposed approach.

Sincerely,

Hiroki Hamaguchi, Naoki Marumo, and Akiko Takeda